

draft-ietf-quic-extended-key- update-03

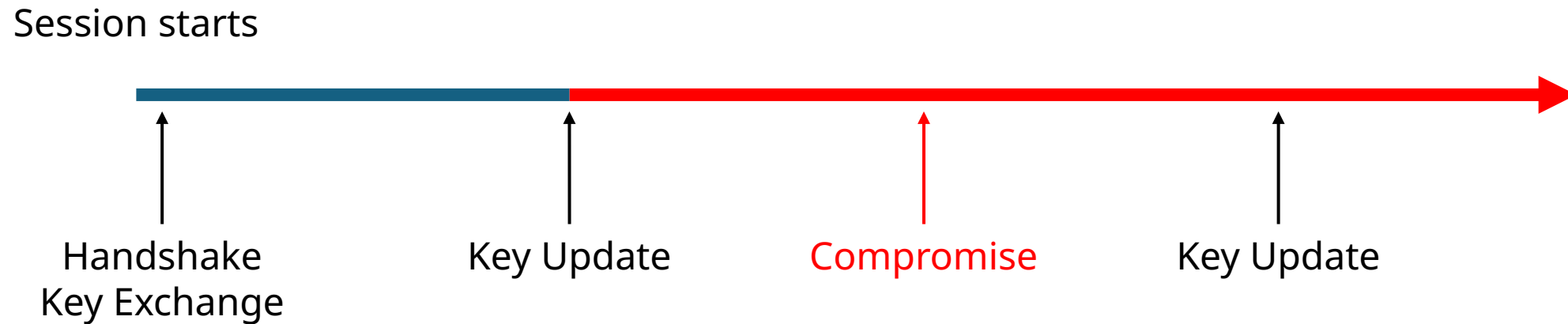
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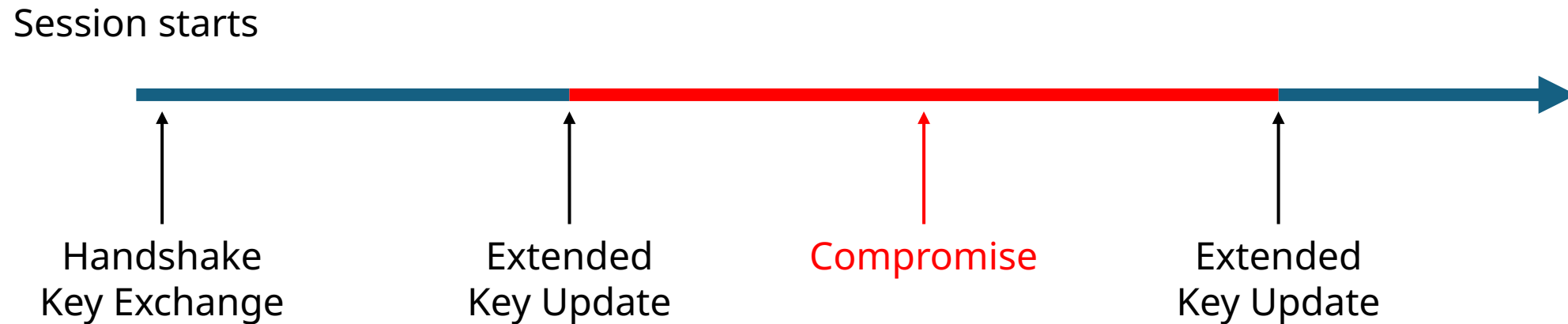
QUIC
IETF126, July 2026, Vienna

Recap: Standard Key Update Security



Attacker can decrypt all packets since the previous Key Update until the end of the session

Recap: **Extended** Key Update Security



Attacker can decrypt only packets between Extended Key Updates

What's happening with the draft

- QUIC ECU depends on TLS ECU
- TLS ECU is going through formal analysis and expected to go through WGLC soon(-ish)
- We have two prototype TLS ECU implementations (mbedTLS and rustls). QUIC (based on rustls/quinn) is in the works

Recent changes in TLS ECU: converge to a single message

ExtendedKeyUpdate (EQU) with uint8 message subtype.

Three subtypes defined:

- key_update_request (0)
- key_update_response (1)
- new_key_update (2) – cannot be used in QUIC

IANA registry will be requested for potential future extensions

Recent changes in TLS ECU: no more failures

- Extended Key Update is always expected to (eventually) succeed.
- No more error codes.
- If responder is too busy, it MAY delay response
 - If requester runs out of patience it terminates the connection

Recent changes in TLS ECU: transcript

- Extended Key Update now carries continuous transcript of all handshake messages
- This is based on recommendation from TLS FATT

Next Steps

- TLS ECU WGLC
- QUIC ECU implementations

Thank you!